Report Maersk Developer Oil Rig

Bürli Norman, Frongillo Matteo, Spichtig David



HS26 – Corporate Ethics and Sustainability – Prof. Dr. Claas Wagner

Q1: Describe your initial operating strategy, covering the following points:
- Production volume
- Overhauls
- Maintenance and safety spending
- Crew management

Initial strategy

The group's initial strategy prioritizes safety while seeking profitable production, combining planned overhauls, maintenance spending, and crew development.

Production volume

In practice round 1 (PR1), the production volume was tested by increasing the production to its maximum (1,650,000 bbls) and hiring more workers. This allowed the group to understand how extreme conditions affect the sustainability factors.
The takeaway for PR1 was that increasing the production volume to its maximum value was not the correct strategy under the specific starting conditions of PR1, as profit was reduced by high hiring costs.

In practice round 2 (PR2), the focus shifted from maximizing crude oil production to testing a more safety-oriented approach.

Decisions in PR1 and PR2 were made to understand how the economic factors respond to these decisions.

The production volume in Round 1 was set to 1,300,000 bbls. To ensure a high volume, ten workers were hired. The same strategy was applied for Round 2.

After Round 1 and Round 2, the group was breaking even. This result is helpful in identifying the threshold at which production becomes profitable under specific conditions.

The next steps in production aim to keep the company profitable.

Overhauls

The overhaul schedule in Figure 1 illustrates how the timing of major overhauls affects the BOP (Blow out preventer) value over successive rounds.

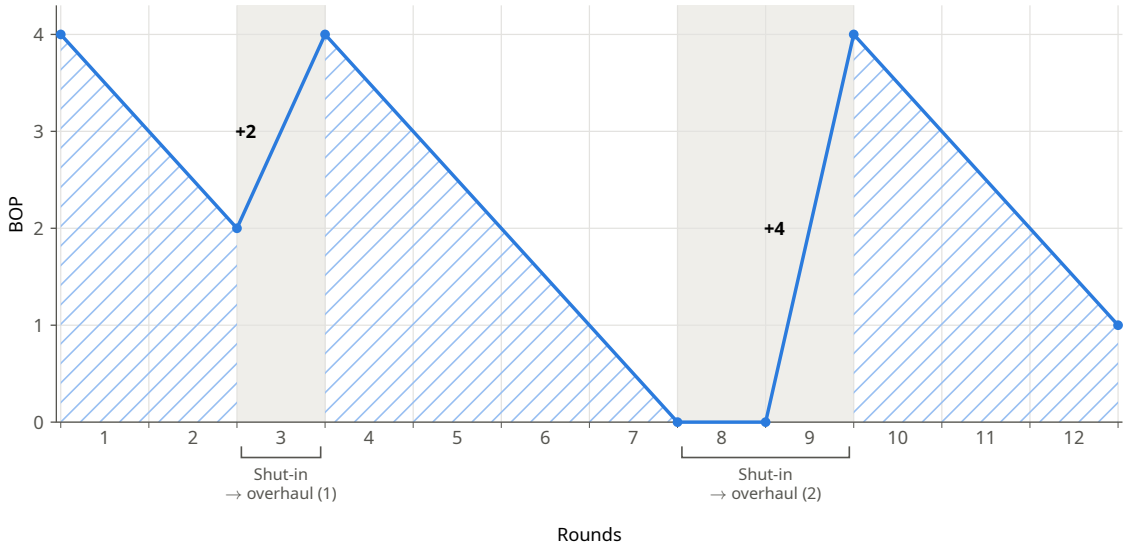


Figure 1: BOP planning schedule

The weather report predicts Beaufort 10 (Storm) in Round 3. A shut-in is necessary if a storm occurs, as the platform and the oil pumps might be compromised. An early overhaul is the best option because it restores the BOP before the expected storm, reducing the risk of a blowout. Assuming that a Beaufort 10 will occur only in Round 3, this strategy allows crude oil production to continue for multiple rounds, meaning that major overhauls would only be needed in Rounds 8 and 9.

The overhaul schedule aims to keep the BOP value above 0. The risk of a blowout exists regardless of the BOP value, but once it reaches 0, the BOP can no longer effectively prevent a blowout.

Maintenance and safety spending

Regular maintenance spending aims to keep the Rig Equipment Condition Index (RECI) above its baseline value of 100.

Two spending approaches are being considered:

1. Invest in maintenance and safety to keep the probability of failures and accidents low
   → Higher spending, lower risk

Table 1: Maintenance and safety expenditure

	Maintenance [$/bbl.]	Safety [$]	RECI
Baseline	8	1,200,000	100
Round 1	12	2,000,000	141
Round 2	12	2,000,000	182

2. Let the RECI degrade and restore it to its baseline value (100) during major overhauls perfomed every six rounds.
   → Cheap but risky.

During Rounds 1 and 2, the first strategy was used. To decide which strategy to adopt, a better understanding of the RECI and its effect on safety is needed.

Crew management

Five crew members were hired in Round 1 to support increased production while keeping the tiredness index low.

Each round, five crew members receive training, allowing expertise to develop over time while keeping training costs low.

Ethical Questions

#1: Arctic Investment Opportunity

The Arctic investment was considered too risky given the expected return and the high initial cost. Therefore, no investment was made.

#2: Backup Battery

The BOP backup battery test costs approximately $5 million.

Table 2: Cost and probability comparison of a blowout and the BOP backup battery test

	Cost [1e6 $]	Estimated chance [%]	Normalized cost [1e6 $]
Blowout (minimum)	250	1	2.5
Backup Battery test of BOP	5	100	5
Blowout (realistic)	500	1	5

A blowout has an estimated minimum cost of $250 million, while a more realistic estimate is $500 million. Assuming a 1% probability of blowout, the expected cost would be $5 million, equal to the cost of the test (Table 2).

Under these assumptions, the test costs as much as the estimated expected loss from a blowout. Carrying out the test is justified as a precaution to verify the backup system's reliability before relying on it during production.

Q2: Discuss the voluntary industry pollution control initiative in R4.

- Explain your original decision and the arguments supporting each option.
- How would egoism evaluate this decision?
- How would utilitarianism evaluate this decision?
- Do these theories support the same conclusion? If not, which one do you find more convincing?

Q3: Analyze strategy-evolution during the simulation.
- Explain any major strategy change and the decision process behind. (consider all points from question 1.)
- What events or observations triggered the change?
- Was the change successful in retrospect?

Q4: Make notes about how you address the missing RECI and expense reduction.

Q5: Discuss the voluntary production cap negotiations in R9.

- Explain your position during the negotiations and the outcome you wanted to achieve.
- Discuss incentives to cooperate and to defect. Why can it be difficult for competing companies to cooperate even when it benefits everyone?
- Why did the eventual outcome occur, and how do you assess your team's contribution?

Q6: Consider the Waste Oil Holding Tank (R8)* and Crude Oil Assay (R10) challenges.

- Explain your reasoning in the two challenges.
- Who could be harmed by each decision, and what obligations did your company have toward them?
- Use Carroll's CSR framework to explain the company's different levels of responsibility that you identify in these two situations.

** Even if you were unable to submit your decision in that round. In that case, consider what your decision would have been.*

Q7: Analyze your production and overhaul decisions during unfavorable weather conditions in R8–11.
- What factors determined whether you attempted production?
- How, if at all, did your willingness to accept risk change during these rounds?
- Do you consider the risks you accepted justified based on the information available at the time?
- To what extent should the actual outcomes (e.g. whether accidents or blowouts eventually occurred) influence your assessment of those decisions?

Q8: Compare your decision-making during the two BOP testing challenges in R2 (Backup Battery) and R11 (Subsea Electronic Module).

- How different did you perceive the two situations to be?
- What were the main arguments in each case?
- How, if at all, did your risk perception or tolerance change?

Q9: Assuming the simulation continues beyond R12, formulate your operating strategy for the next phase. Identify the most important learnings from R1–R12 underlying this strategy.

Appendix

List of Figures

1 BOP planning schedule 2

List of Tables

1 Maintenance and safety expenditure 3
2 Cost and probability comparison of a blowout and the BOP backup battery test 3

Declarations on the use of AI tools

"*ChatGPT 6*" was used to enhance vocabulary.
All sentences originate from our own ideas and were refined with the support of this tool.
https://chatgpt.com/